

COMP1017 (Mathematics for Computer Scientists) Test 1

Academic Session 2017-2018, Autumn Semester

There are TWO (2) questions pertaining to propositional logic and predicate logic on TWO (2) pages. Please answer ALL the questions. Time allowed 30 minutes.

1. Let p be the proposition “He will come on the 8:15 train”, q be “He will come on the 9:15 train”, and r be “He will have time to visit us”. Given the following propositional formulas that

$$(1) \quad (p \vee q) \leftrightarrow r$$

$$(2) \quad (p \rightarrow r) \wedge (p \vee r)$$

$$(3) \quad (p \vee \neg q) \rightarrow r$$

$$(4) \quad (q \leftrightarrow r) \vee (q \leftrightarrow \neg r)$$

(a) Find the truth table for (1).

[7 marks]

(b) Show with the logical equivalence laws that (2) can be simplified to r .

[8 marks]

(c) Find the contrapositive for (3) and express it in simple English.

[8 marks]

(d) Show that (4) is a tautology.

[11 marks]

(e) Express the sentence 'He will come on the 8:15 or the 9:15 train, and if the former, he will have time to visit us' as the propositional formula with p , q , r , and logical operators.

[8 marks]

2. Let $G(x, y)$ be the statement “ x gets y ” and $W(x, y)$ be the statement “ x wants y ”, respectively. Let the universe of discourse of x be all people, and let the universe of discourse of y be all things that are other than people.

(a) Translate the following sentences into predicate logic notation using $G(x, y)$, $W(x, y)$, and quantifiers.

(i) If everyone wants something, then no one gets it.

[12 marks]

(ii) Some people want only things they don't get.

[12 marks]

(iii) There is something that nobody wants.

[8 marks]

(b) If we let the universe of discourse of both x and y be all things and all people, we will have to introduce an additional predicate, in order to translate the sentences in (a) into predicate logic notation. What should it be? Explain your work.

[8 marks]

(c) How many free variables are there in each of the following predicate logic expressions? Explain your answers.

(i) $(\forall x G(x, y)) \wedge (\exists y W(x, y))$

[6 marks]

(ii) $\exists x, y (W(x, y) \rightarrow G(x, t))$

[6 marks]

(iii) $\exists s, t \forall x, y (W(s, y) \leftrightarrow G(x, t))$

[6 marks]